

Section 1:

EdgeImpulse link: <https://studio.edgeimpulse.com/public/1074074/live>

1.

The model uses three accelerometer axes:

- accX
- accY
- accZ

The sampling frequency is 100 Hz, with a 2000 ms window and 220 ms stride.

2.

The NN classifier receives spectral features generated from the three accelerometer axes. It uses 33 input features. Five example feature names are:

- accX RMS
- accX Peak 1 Height
- accY Peak 3 Height
- accX Peak 1 Freq
- accY RMS

The network contains dense layers with 32 and 16 neurons before the two-class output layer.

3.

The classifier outputs two classes:

- nominal
- off

4.

The anomaly detector uses 4 selected spectral features:

- accX RMS
- accX Peak 1 Height
- accY Spectral Power 2.0–5.0 Hz
- accZ Spectral Power 2.0–5.0 Hz

It uses 32 k-means clusters and produces one anomaly score.

5.

```

nominal: 0.437500
off: 0.562500
Anomaly prediction: 16.023445
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 44 ms, inference 610 us, anomaly 6 ms, postprocessing 49 us
#Classification predictions:
  nominal: 0.437500
  off: 0.562500
Anomaly prediction: 16.025091
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 44 ms, inference 593 us, anomaly 6 ms, postprocessing 50 us
#Classification predictions:
  nominal: 0.437500
  off: 0.562500
Anomaly prediction: 16.024462
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 44 ms, inference 595 us, anomaly 6 ms, postprocessing 49 us
#Classification predictions:
  nominal: 0.437500
  off: 0.562500
Anomaly prediction: 16.025341
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 44 ms, inference 593 us, anomaly 6 ms, postprocessing 49 us
#Classification predictions:
  nominal: 0.437500
  off: 0.562500
Anomaly prediction: 16.024240
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 44 ms, inference 591 us, anomaly 6 ms, postprocessing 50 us
#Classification predictions:
  nominal: 0.402344
  off: 0.597656
Anomaly prediction: 15.994625
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 44 ms, inference 593 us, anomaly 6 ms, postprocessing 49 us
#Classification predictions:
  nominal: 0.437500
  off: 0.562500
Anomaly prediction: 16.004496
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 44 ms, inference 611 us, anomaly 6 ms, postprocessing 49 us
#Classification predictions:
  nominal: 0.437500
  off: 0.562500
Anomaly prediction: 16.003010
Starting inferencing in 2 seconds...

```

6.

The Edge Impulse model classified the input as off, with approximately 56 - 60% confidence, while the nominal probability remained around 40 - 44%. The anomaly score stayed close to 16 across repeated measurements, indicating that the observed sensor data was far from the patterns represented by the anomaly detector's training clusters. Because the class probabilities were relatively close and the anomaly score was high, the classifier output should be treated with caution. A high anomaly score suggests that the input may not resemble the training data closely enough for the class prediction to be reliable.

Section 2:

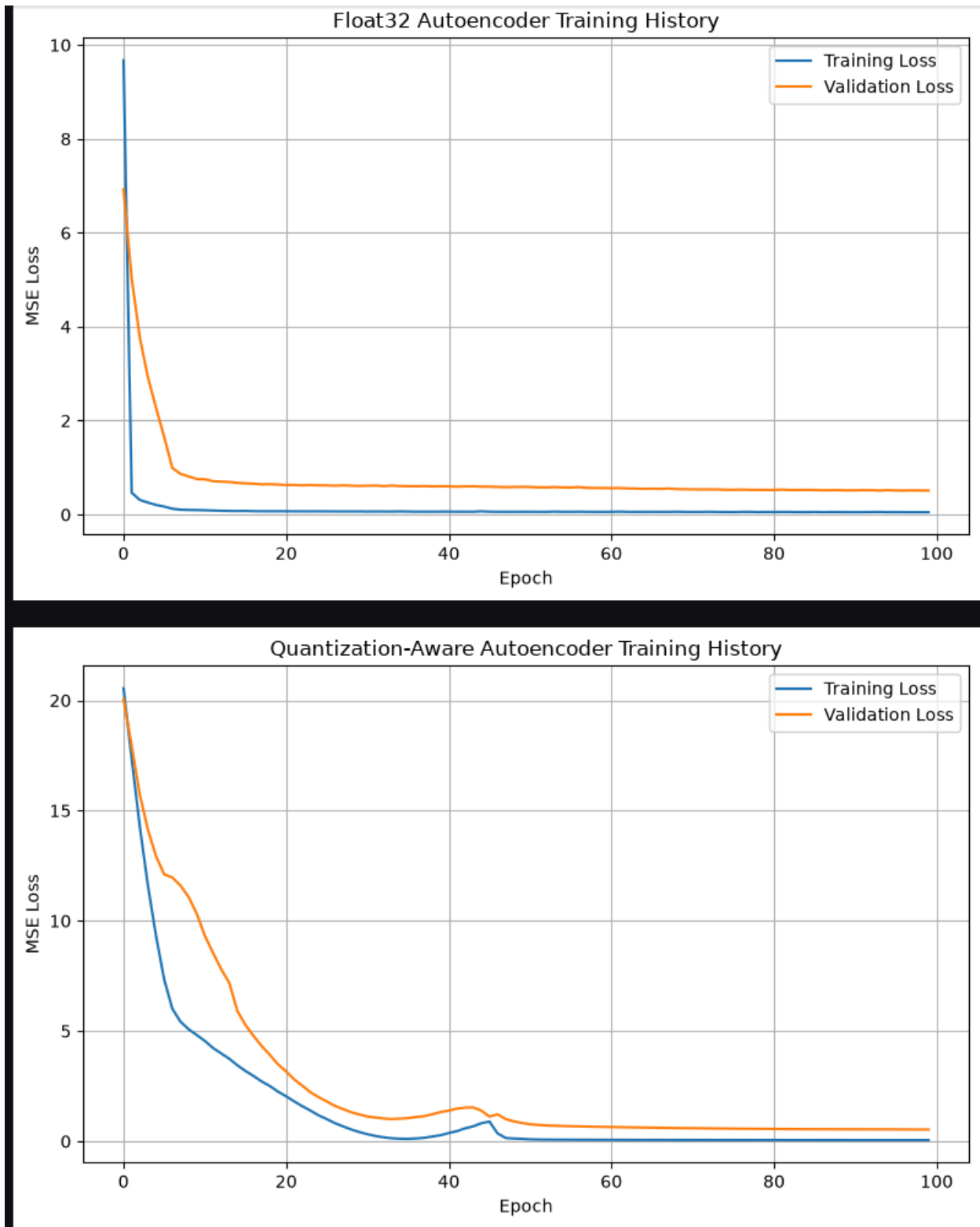


Figure 1. Float32 and quantization-aware autoencoder training and validation loss across epochs.

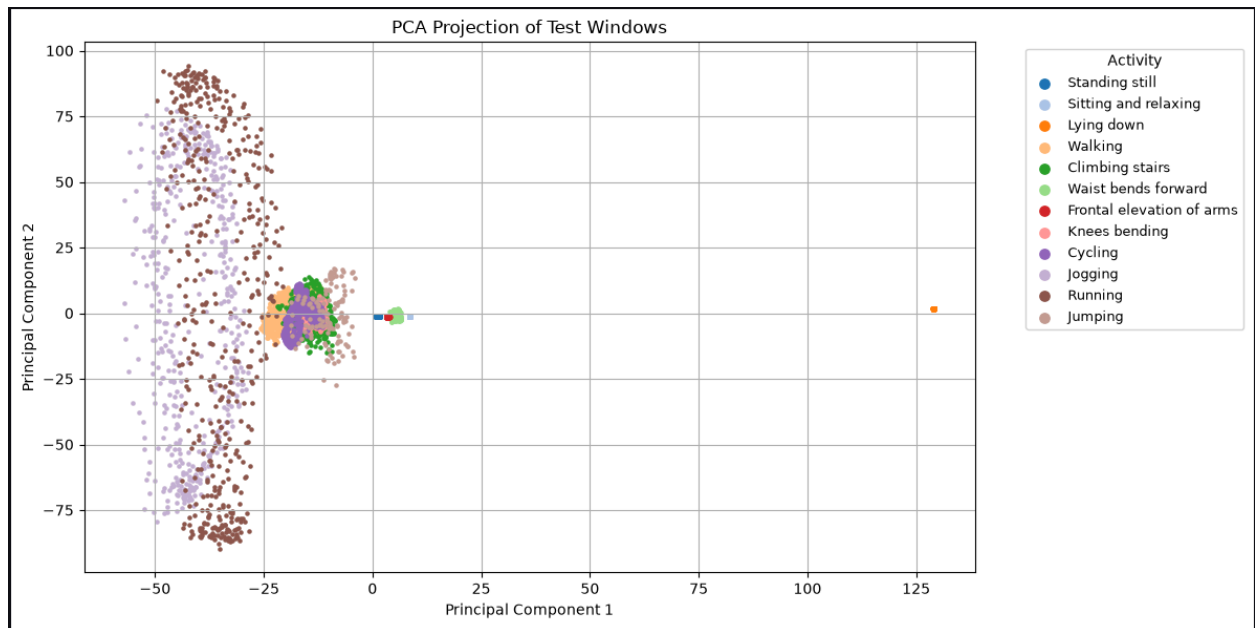


Figure 2. PCA projection of the activity windows into two dimensions, showing the distribution of normal and anomalous samples.

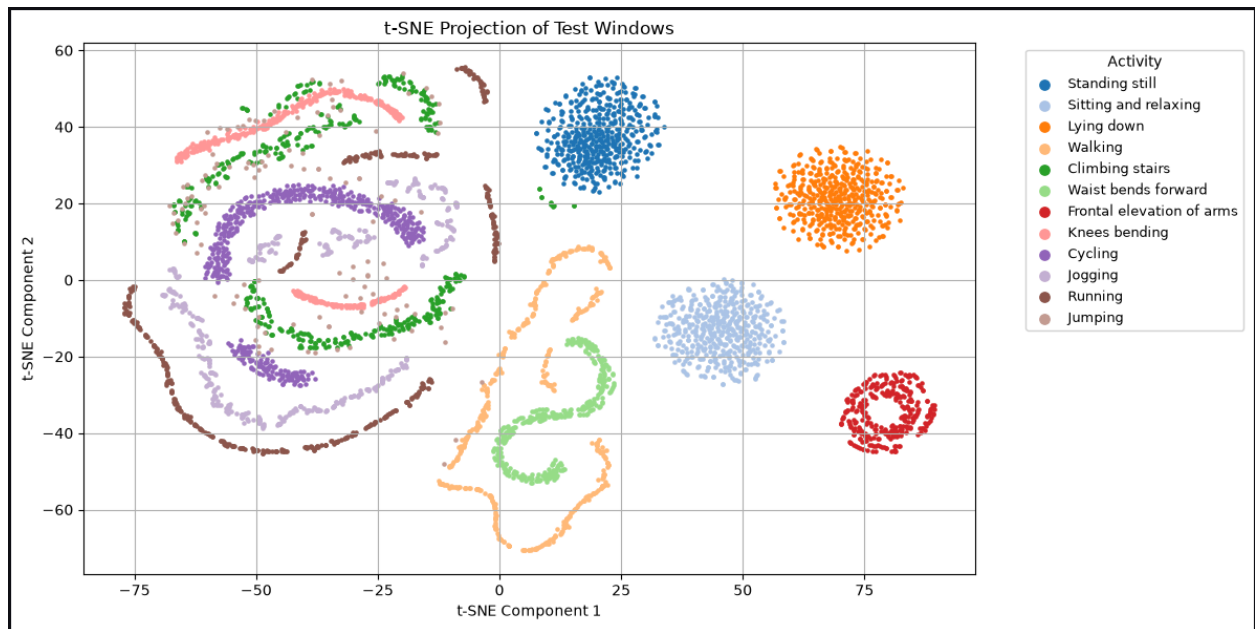


Figure 3. t-SNE projection of the activity windows into two dimensions, showing local clustering patterns among the activity samples.

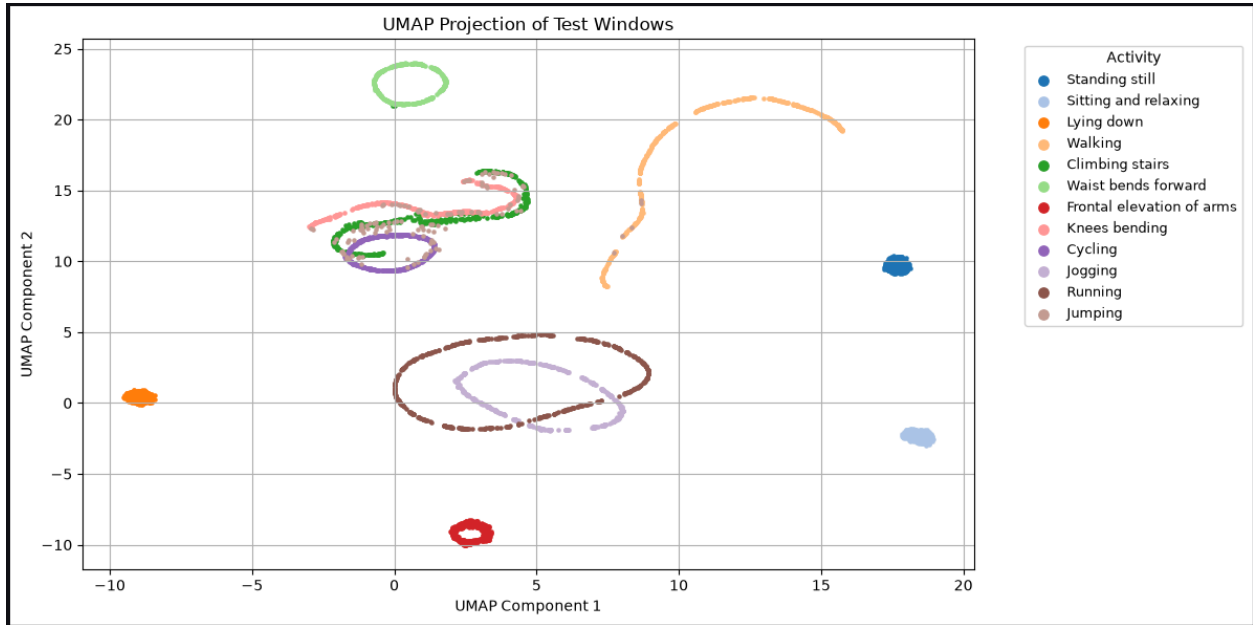


Figure 4. UMAP projection of the activity windows into two dimensions, showing the separation and structure of normal and anomalous samples.

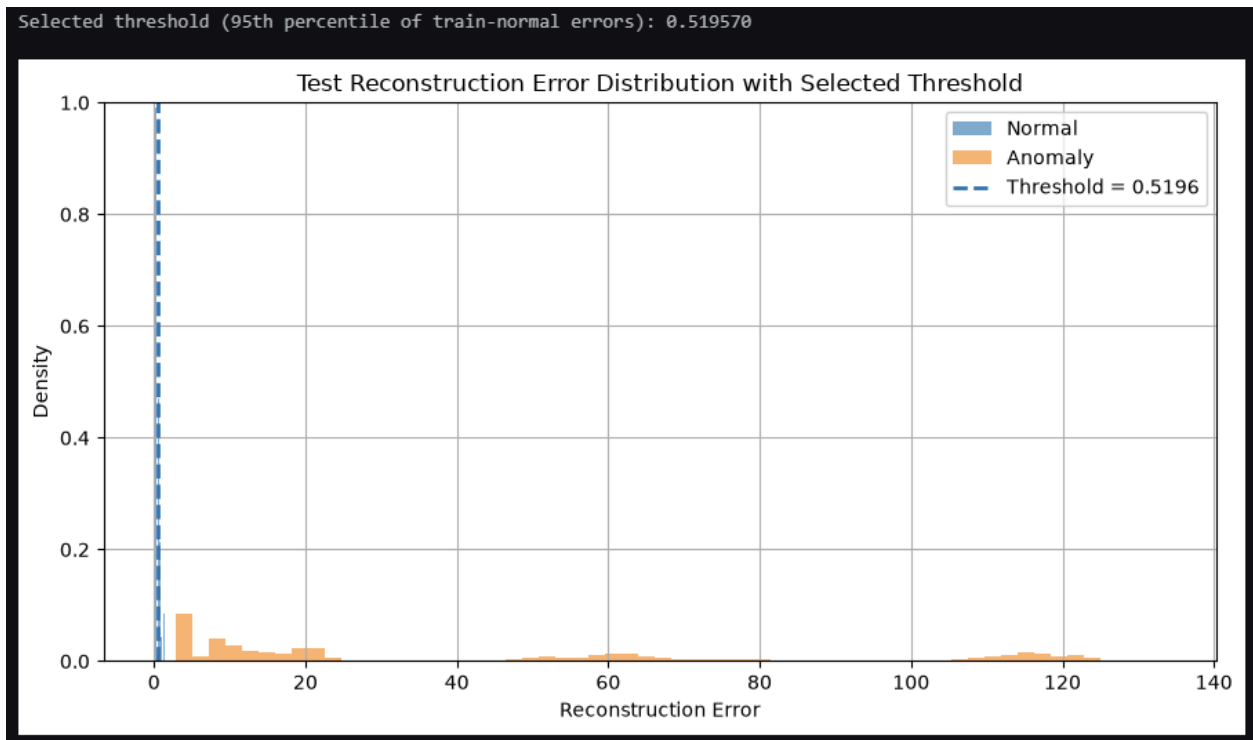


Figure 5. Test reconstruction error distributions for normal and anomalous samples, with the selected anomaly threshold of 0.519570 shown by the dashed vertical line.

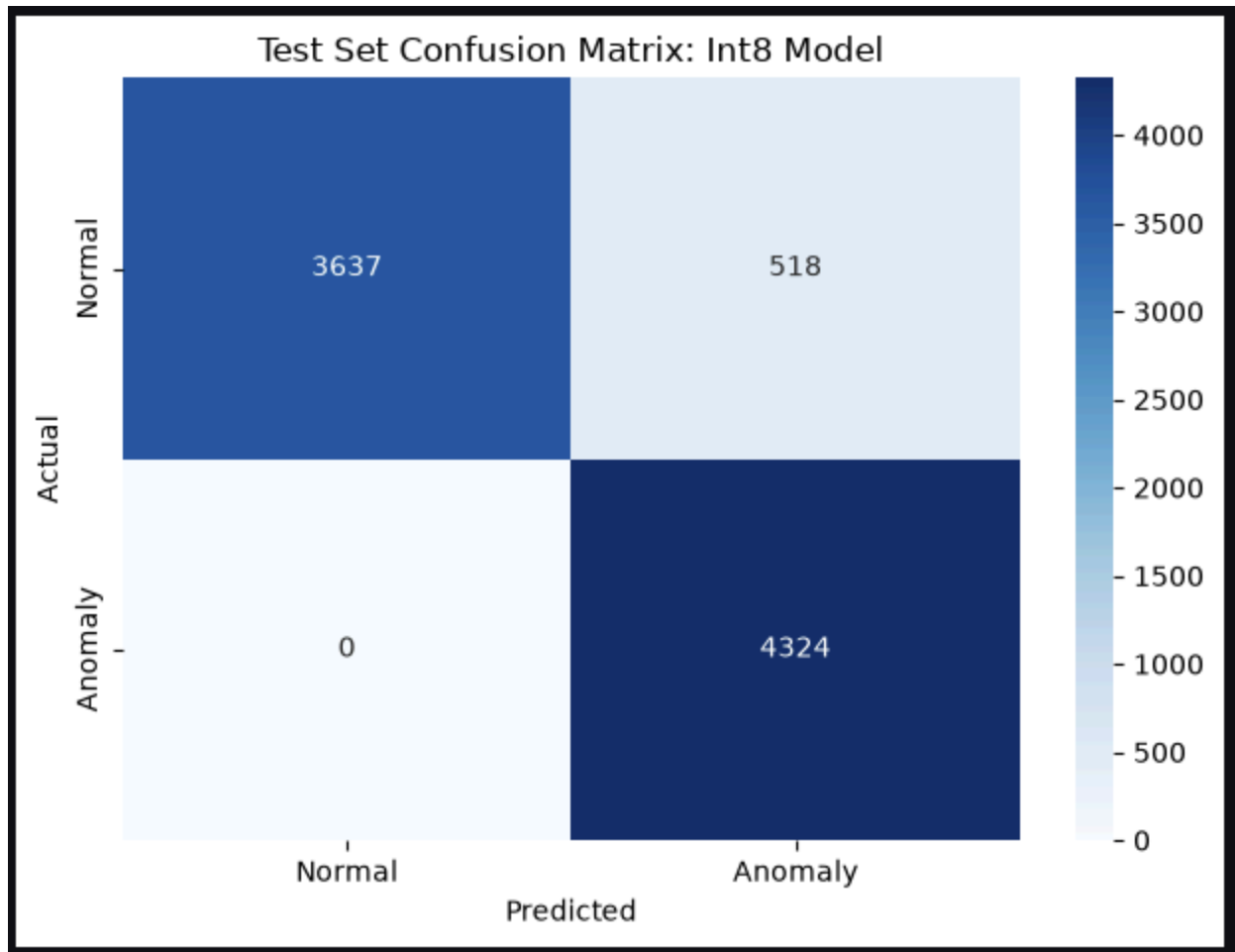


Figure 6. Test-set confusion matrix for the quantized int8 autoencoder anomaly detection model.

Train set evaluation				
	precision	recall	f1-score	support
Normal (0)	1.00	0.95	0.97	10478
Anomaly (1)	0.95	1.00	0.98	10872
accuracy			0.98	21350
macro avg	0.98	0.97	0.98	21350
weighted avg	0.98	0.98	0.98	21350
Test set evaluation				
	precision	recall	f1-score	support
Normal (0)	1.00	0.88	0.93	4155
Anomaly (1)	0.89	1.00	0.94	4324
accuracy			0.94	8479
macro avg	0.95	0.94	0.94	8479
weighted avg	0.95	0.94	0.94	8479

Figure 7. Train- and test-set classification metrics for the quantized int8 autoencoder, including precision, recall, F1-score, and accuracy.

```
Result: Normal activity
Reconstruction error: 0.728996
Result: Normal activity
Reconstruction error: 0.730888
Result: Normal activity
Reconstruction error: 0.782235
Result: Normal activity
Reconstruction error: 0.878065
Result: Normal activity
Reconstruction error: 0.926038
Result: Normal activity
Reconstruction error: 0.969644
Result: Normal activity
Reconstruction error: 0.960886
Result: Normal activity
Reconstruction error: 0.978457
Result: Normal activity
Reconstruction error: 1.025300
Result: Normal activity
Reconstruction error: 1.089224
Result: Normal activity
Reconstruction error: 1.167991
Result: Normal activity
Reconstruction error: 1.266963
Result: Normal activity
Reconstruction error: 1.297114
Result: Normal activity
Reconstruction error: 1.375373
Result: Normal activity
Reconstruction error: 1.388560
Result: Normal activity
Reconstruction error: 1.537571
Result: Anomaly detected
Reconstruction error: 1.597925
Result: Anomaly detected
Reconstruction error: 1.663737
Result: Anomaly detected
Reconstruction error: 1.610198
Result: Anomaly detected
Reconstruction error: 1.626972
Result: Anomaly detected
Reconstruction error: 1.650354
Result: Anomaly detected
Reconstruction error: 1.715582
Result: Anomaly detected
Reconstruction error: 1.778646
Result: Anomaly detected
Reconstruction error: 1.968946
Result: Anomaly detected
Reconstruction error: 2.096824
Result: Anomaly detected
Reconstruction error: 2.296806
Result: Anomaly detected
Reconstruction error: 2.278544
Result: Anomaly detected
Reconstruction error: 2.356116
Result: Anomaly detected
Reconstruction error: 2.417472
Result: Anomaly detected
Reconstruction error: 2.528027
Result: Anomaly detected
Reconstruction error: 2.528561
Result: Anomaly detected
```

Figure 8. Arduino Nano 33 BLE Sense serial monitor output showing reconstruction error values and normal/Anomaly classification results.

1.

I used a reconstruction-error threshold of 0.519570, selected as the 95th percentile of the reconstruction errors from normal training samples. This allows most normal activity to remain below the threshold while flagging unusually high errors as anomalies.

2.

The autoencoder is trained mainly on normal activity, so it reconstructs normal inputs accurately and produces a low error. Anomalous activities are less familiar to the model, so they are reconstructed poorly and produce a higher error.

3.

The deployed system must preserve the quantized model bytes, input size and window length, selected sensor channels, normalization or scaling values, int8 quantization parameters, reconstruction-error calculation, and the anomaly threshold.

4.

The Arduino must process sensor data exactly as the notebook did during training. Different window sizes, channel order, scaling, or normalization would change the model input distribution and could cause incorrect reconstruction errors and classifications.

5.

The method is limited by the Arduino's memory, processing speed, and power. It may also produce false positives when normal behavior changes, fail to recognize anomalies similar to normal activity, and require retraining if the sensor placement, user, or environment changes.